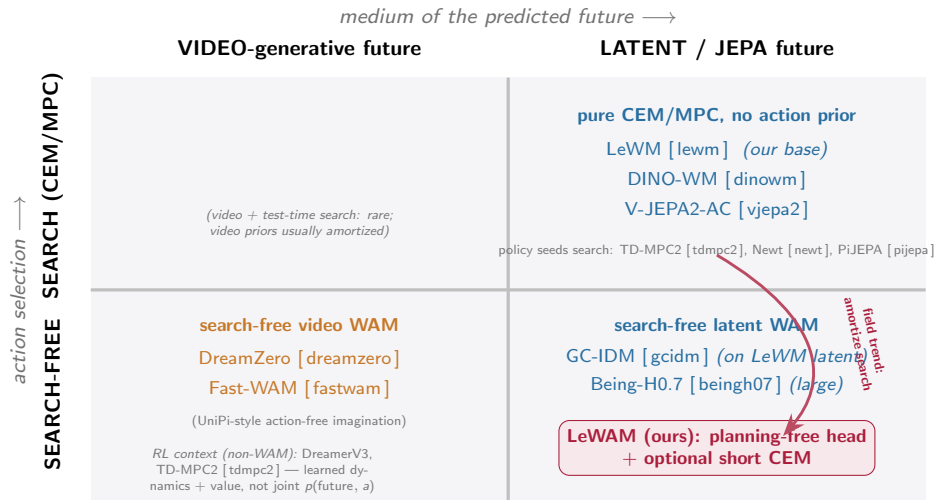


LeWAM: A Simple, Fast, Scalable World–Action Model

Related work + our goal

The world-action model (WAM) landscape



Map. A WAM jointly models $p(\text{future}, a)$. Axes: future medium (video↔latent) × action selection (test-time search↔amortized one forward pass). The field is converging toward **search-free**. LeWAM is a **latent** WAM built on LeWM, planning-free by default with optional short search.

Related work I — video-generative WAMs

The future is a **video**: a generative model predicts pixels, actions come from the imagined rollout.

- **DreamZero** [dreamzero] — video diffusion world model + inverse-dynamics action read-out; search-free at test time.
- **Being-H0.7** [beingh07] — large-scale latent/video world-action model trained from egocentric video.
- **UniPi** [unipi] — text-conditioned video generation as a universal policy; inverse dynamics maps frames to actions.

Strength. Search-free action read-out; scales with web-scale video and pretrained generators.

Cost. Heavy generative rollouts (full-frame diffusion per step); large data and compute budgets; pixel prediction spends capacity on visual detail irrelevant to control.

Related work II — latent WAMs with test-time search

The future is a **latent**: a JEPA-style predictor rolls forward in embedding space; actions are chosen by planning (CEM/MPC) over that predictor.

- **LeWM** [lewm] (*our base*) — stable end-to-end JEPA world model from pixels; learnable encoder, action-conditioned latent predictor.
- **DINO-WM** [dinowm] — world model on frozen DINO features; zero-shot planning by latent rollout.
- **V-JEPA2-AC** [vjepa2] — action-conditioned video JEPA; latent prediction for planning.

Strength. Light latent prediction (no pixel decode); strong goal-reaching by optimizing actions against the predictor.

Gap. No learned action prior. Every step re-samples actions *blind* (CEM population $\times$ horizon $\times$ iterations), paying the full search cost at deployment.

Related work III — search-free / amortized policies

Amortize the planner into a network: one forward pass replaces test-time search.

- **GC-IDM** [gcidm] — goal-conditioned inverse dynamics on the LeWM latent; $\sim 100\text{--}130\times$ cheaper than CEM. *Markovian: single-frame* (z_t, z_g, h).
- **Fast-WAM** [fastwam] — argues a world-action model needs *no* test-time future imagination.
- Diffusion-policy context — **Diffuser** [diffuser], **Decision-Diffuser** [decisiondiffuser], **BESO** [beso]: generative trajectory/action heads, no latent-WM rollout.

Strength. One forward pass per action; deployment-cheap.

Note. GC-IDM conditions on a *single* frame (Markovian); diffusion heads add K denoising steps and do not reuse a world-model predictor.

Previous work vs LeWAM

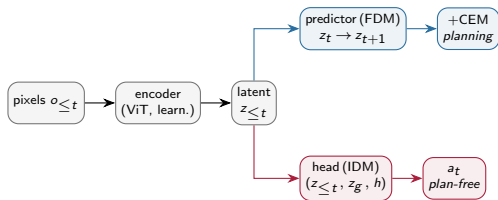
Prior family	What they do / limit	What LeWAM does differently
Pure-CEM latent WMs LeWM [lewm], DINO-WM [dinowm], V-JEPA2-AC [vjepa2]	Search <i>blind</i> at every step; no learned action prior.	Adds a planning-free action head $\Rightarrow$ $\sim 1104\times$ fewer FLOPs/action (measured); keep search only where it helps.
Amortized GC-IDM GC-IDM [gcidm]	One forward pass, but <i>Markovian</i> (single frame); one control mode.	History-conditioned head; one model unifies three settings (planning / goal-conditioned / BC).
Video-generative WAMs DreamZero [dreamzero], Being-H0.7 [beingh07]	Heavy generative rollouts; large data/compute.	Light vit-tiny latent ; no pixel decode, no generative rollout.

Honest positioning. LeWAM's contribution is the **efficiency Pareto**, the **multi-setting per-family analysis**, and being built on LeWM [lewm] — *not* a state-of-the-art success rate.

LeWAM = LeWM latent + a planning-free action head

One trained model, three control settings.

- **Base:** LeWM [lewm] — stable end-to-end JEPA world model from pixels (learnable encoder, action-conditioned predictor).
- **Ours:** a history/goal/horizon-conditioned **action head** trained jointly on the same latent.
- Decodes into 3 settings without retraining: *planning* (predictor + CEM), *goal-conditioned policy*, *behavior cloning*.



blue = forward dynamics $\rightarrow$ search red = action head $\rightarrow$ one pass

What is ours (honest scope): the action head, the efficiency result, and the per-setting analysis. The base WM is LeWM. No SOTA success-rate claim.

Goal. A *Simple, Fast, Scalable* WAM. The contribution is the **efficiency Pareto** — amortized action $\approx$ search at a fraction of the cost — plus the per-setting analysis, not a higher success rate.

The contribution: the efficiency Pareto (measured)

Planning-free LeWAM decides each action in **one forward pass**. Search (CEM/MPPI) re-rolls a population over a horizon for many iterations; diffusion heads multiply by K denoising steps.

Per-action cost (can, ViT-tiny, fp32, batch 1)

method	GFLOPs/action	vs. CEM
LeWAM-GC (ours)	2.87	1104× fewer
GC-IDM [gcidm]	5.60	565× fewer
DP/LDP head ($K=50$)	3.73	~850× fewer
CEM / MPC [lewm]	3165	1×

CEM blow-up = 300 samples $\times$ horizon 5 $\times$ 30 iters = 45,000 rollouts.

Headline (verified):

- **1104×** fewer FLOPs/action than CEM.
- **~90–132×** faster wall-clock than CEM.
- **~3.5×** faster than a diffusion ($K=50$) head.

Why: the ViT encoder dominates cost; planning-free runs it once, CEM runs the predictor over a 45k-rollout population.

TD-MPC2 plans at test and reports *no* per-action latency — this speed axis is uncontested.

Claim. SR competitive while far cheaper. Keep planning where it helps, fall back to the planning-free head elsewhere. Contribution = efficiency + analysis, not SOTA SR.